Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Processing (01CT1513)	**Aim : To apply low-pass and band-pass filters to an audio signal and observe their effects on sound quality and frequency components.**	
OpenEnded :- 4	**Date: 20/10/2026**	**Enrollment No: 92400133125**

Git Hub : https://github.com/mankumarbhalodiya/Digital-Signal-and-Image-Processing

Q3. (a)

Select any MP3 song of your choice and process it using a low-pass filter that you design. Experiment with at least three different cutoff frequencies and observe how each setting changes the filtered audio output. Discuss your observations in detail, focusing on how the cutoff frequency influences the clarity, tonal balance, and overall character of the song.

Code :

```
import librosa

import numpy as np

import matplotlib.pyplot as plt

import soundfile as sf

from scipy.signal import butter, sosfiltfilt

filename =
"Todh___Prince_Narula___Munawar__Official_Video____Jaymeet___Rony_Ajnali___Gill_Machhrai(128k).mp3"

y, sr = librosa.load(filename, sr=None)

print("Sample Rate:", sr)

print("Duration:", len(y) / sr, "seconds")

def lowpass_filter(data, cutoff, fs, order=6):

    sos = butter(order, cutoff, btype="lowpass", fs=fs, output="sos")

    return sosfiltfilt(sos, data)

cutoffs = [500, 2000, 5000]
```

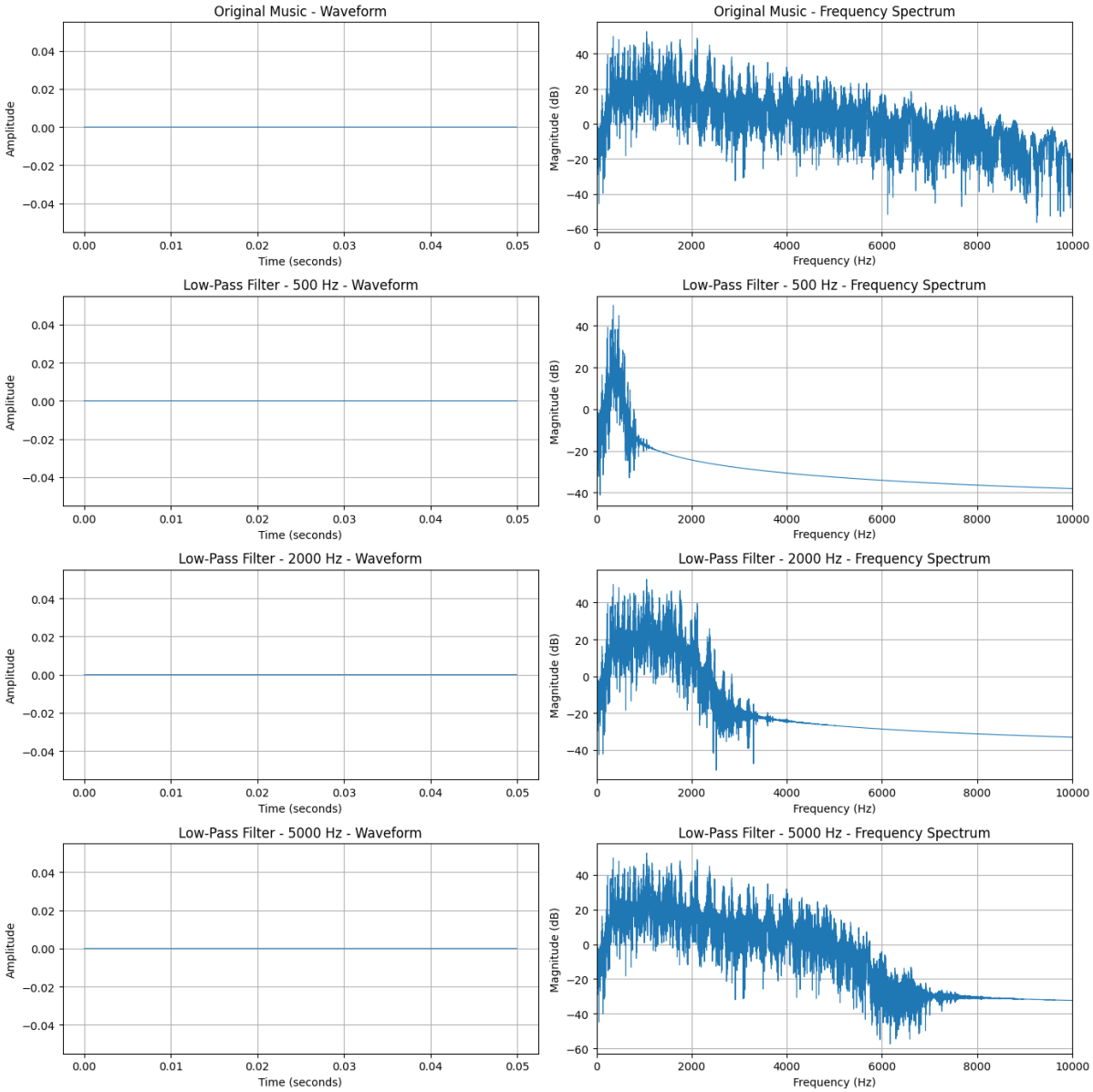
Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Processing (01CT1513)	**Aim : To apply low-pass and band-pass filters to an audio signal and observe their effects on sound quality and frequency components.**	
OpenEnded :- 4	**Date: 20/10/2026**	**Enrollment No: 92400133125**

```
signals = [("Original Music", y)]
for cutoff in cutoffs:
    filtered = lowpass_filter(y, cutoff, sr)
    sf.write(f"lowpass_{cutoff}Hz.wav", filtered, sr)
    signals.append((f"Low-Pass Filter - {cutoff} Hz", filtered))
samples = min(sr // 20, len(y))
time = np.arange(samples) / sr
fig, axes = plt.subplots(4, 2, figsize=(14, 14))
for i, (title, signal) in enumerate(signals):
    segment = signal[:samples]
    axes[i, 0].plot(time, segment, linewidth=0.8)
    axes[i, 0].set_title(title + " - Waveform")
    axes[i, 0].set_xlabel("Time (seconds)")
    axes[i, 0].set_ylabel("Amplitude")
    axes[i, 0].grid(True)
    spectrum = np.abs(np.fft.rfft(signal[:min(len(signal), sr * 5)]))
    frequencies = np.fft.rfftfreq(
        min(len(signal), sr * 5), d=1 / sr
    )
    spectrum_db = 20 * np.log10(spectrum + 1e-10)
    axes[i, 1].plot(frequencies, spectrum_db, linewidth=0.8)
    axes[i, 1].set_title(title + " - Frequency Spectrum")
    axes[i, 1].set_xlabel("Frequency (Hz)")
```

Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Processing (01CT1513)	**Aim : To apply low-pass and band-pass filters to an audio signal and observe their effects on sound quality and frequency components.**	
OpenEnded :- 4	**Date: 20/10/2026**	**Enrollment No: 92400133125**

```
    axes[i, 1].set_ylabel("Magnitude (dB)")
    axes[i, 1].set_xlim(0, 10000)
    axes[i, 1].grid(True)
plt.tight_layout()
plt.show()
print("All filtered audio files saved successfully.")
```

Output :



Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Processing (01CT1513)	**Aim : To apply low-pass and band-pass filters to an audio signal and observe their effects on sound quality and frequency components.**	
OpenEnded :- 4	**Date: 20/10/2026**	**Enrollment No: 92400133125**

(b) Use an appropriate band-pass filter to separately suppress vocals and background music, saving each filtered result as its own MP3 file. Analyze and describe your observations, focusing on how each filter affects the different components of the song.

Code :

```
import librosa

import librosa.display

import numpy as np

import matplotlib.pyplot as plt

import soundfile as sf

from scipy.signal import butter, lfilter

filename =
"Todh___Prince_Narula___Munawar__Official_Video____Jaymeet___Rony_Ajnali___Gill_Ma
chhrai(128k).mp3"

y, sr = librosa.load(filename, sr=None)

print(f"Sample rate: {sr}, Duration: {len(y)/sr:.2f} seconds")

def bandpass_filter(data, lowcut, highcut, fs, order=6):

    nyq = 0.5 * fs

    b, a = butter(order, [lowcut / nyq, highcut / nyq], btype="band")

    return lfilter(b, a, data)

vocal_band = bandpass_filter(y, 300, 3400, sr)

background_band = y - vocal_band

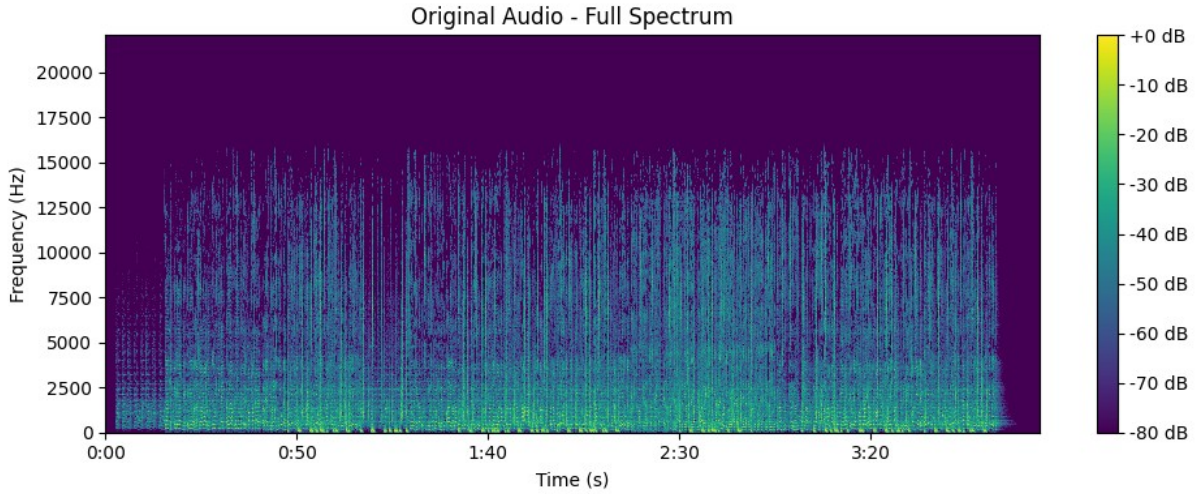
def plot_spectrogram(signal, sr, title):

    D = librosa.amplitude_to_db(np.abs(librosa.stft(signal)), ref=np.max)
```

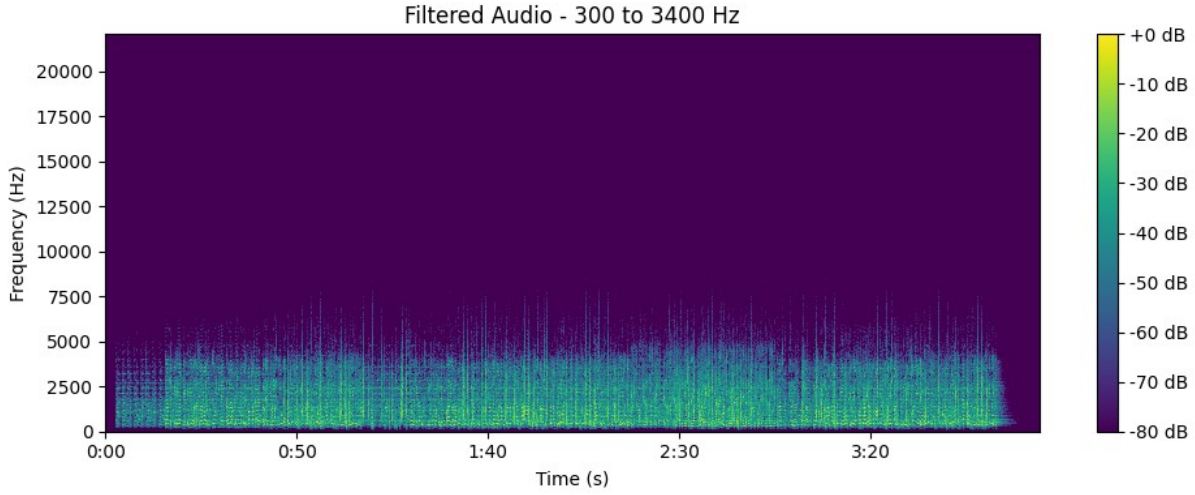
Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Processing (01CT1513)	**Aim : To apply low-pass and band-pass filters to an audio signal and observe their effects on sound quality and frequency components.**	
OpenEnded :- 4	**Date: 20/10/2026**	**Enrollment No: 92400133125**

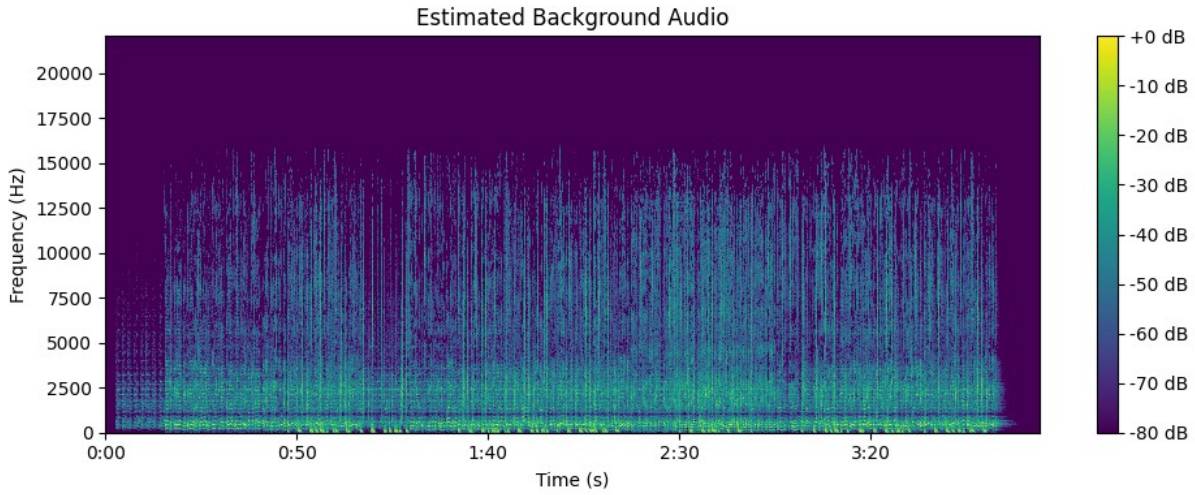
```
    plt.figure(figsize=(10, 4))
    librosa.display.specshow(D, sr=sr, x_axis="time", y_axis="hz")
    plt.colorbar(format="%+2.0f dB")
    plt.title(title)
    plt.xlabel("Time (s)")
    plt.ylabel("Frequency (Hz)")
    plt.tight_layout()
    plt.show()
plot_spectrogram(y, sr, "Original Audio")
plot_spectrogram(vocal_band, sr, "Filtered Audio")
plot_spectrogram(background_band, sr, "Estimated Background Audio")
sf.write("filtered_vocals.wav", vocal_band, sr)
```

Output :



Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Processing (01CT1513)	**Aim : To apply low-pass and band-pass filters to an audio signal and observe their effects on sound quality and frequency components.**	
OpenEnded :- 4	**Date: 20/10/2026**	**Enrollment No: 92400133125**





Conclusion for Q3(a): Low-Pass Filter :

The low-pass filter was applied to the original audio using three different cutoff frequencies. I observed that a lower cutoff frequency removes more high-frequency components, making the audio sound duller, while a higher cutoff frequency preserves more of the original sound. The waveform and frequency spectrum helped me understand how the filter changes the audio signal. Overall, this experiment showed how the cutoff frequency affects the clarity and tonal quality of music.

Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Processing (01CT1513)	**Aim : To apply low-pass and band-pass filters to an audio signal and observe their effects on sound quality and frequency components.**	
OpenEnded :- 4	**Date: 20/10/2026**	**Enrollment No: 92400133125**

Conclusion for Q3(b): Band-Pass Filter :

The band-pass filter was used to keep a selected range of frequencies while reducing frequencies outside that range. I observed that changing the frequency range affects the sound quality and the amount of vocals and background music that can be heard. Although the filter can make certain parts of the audio more noticeable, it cannot completely separate vocals from background music. This experiment helped me understand how band-pass filtering can be used to analyze and modify audio signals.